



Global Positioning System

Space Segment



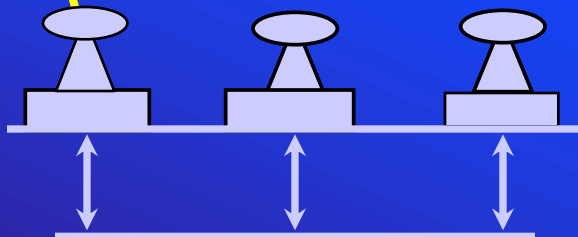
Broadcast of:

- Navigation Data
- Spread-spectrum ranging signals

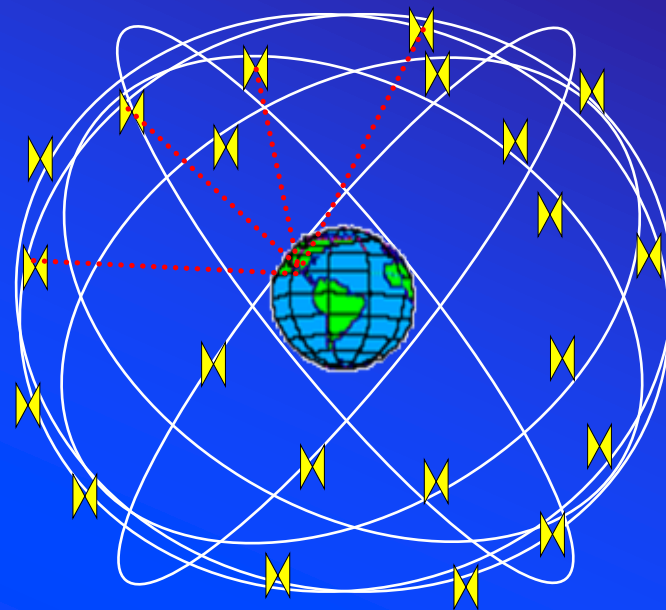
Uplink of:

- Ephemeris
- Clock Data

Control Segment



Five Monitor Stations & Four Upload Stations Spread Around the Equator



User Segment



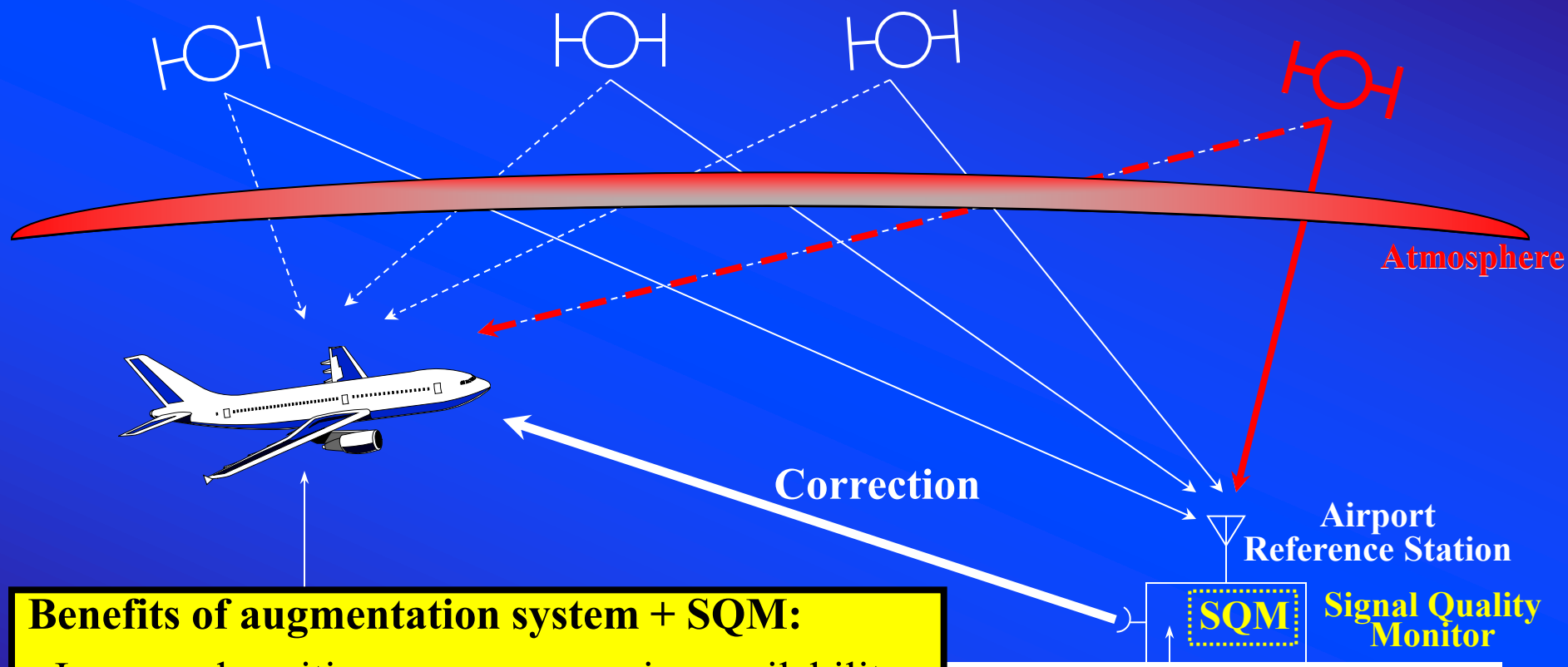
Requirements For Aviation

	Standalone GPS	Precision Approach	
		Cat. I (DH 200 ft.)	Cat. III (DH 50 ft.)
Accuracy (95%)	H: 10 meters V: 20 meters	H: 16 meters V: 4.0 meters	H: 6.1 meters V: 2.0 meters
Alarm level (Prob. MI)	Unspecified	(2×10^{-7} / approach) H: 40 meters V: 10 meters	(2×10^{-9} / approach) H: 15.5 meters V: 5.3 meters
Time to alarm	Unspecified	≤ 6 seconds	≤ 2 seconds

- Standalone GPS does not meet the requirements for precision approach (vertical accuracy insufficient; integrity not guaranteed).
- To meet the requirements, GPS is augmented by additional ground-based equipment.



Local Area Augmentation System



Benefits of augmentation system + SQM:

- Improved position accuracy, service availability
- Tighter error bounds
- Continuous monitoring of satellites
- Some architectures also improve geometry

Known:

E: 122°47'25"

N: 37° 10'39"

U: 22.57m

Received:

E: 122°

N: 37°

U: 31m



GPS Signal Structure

Carrier

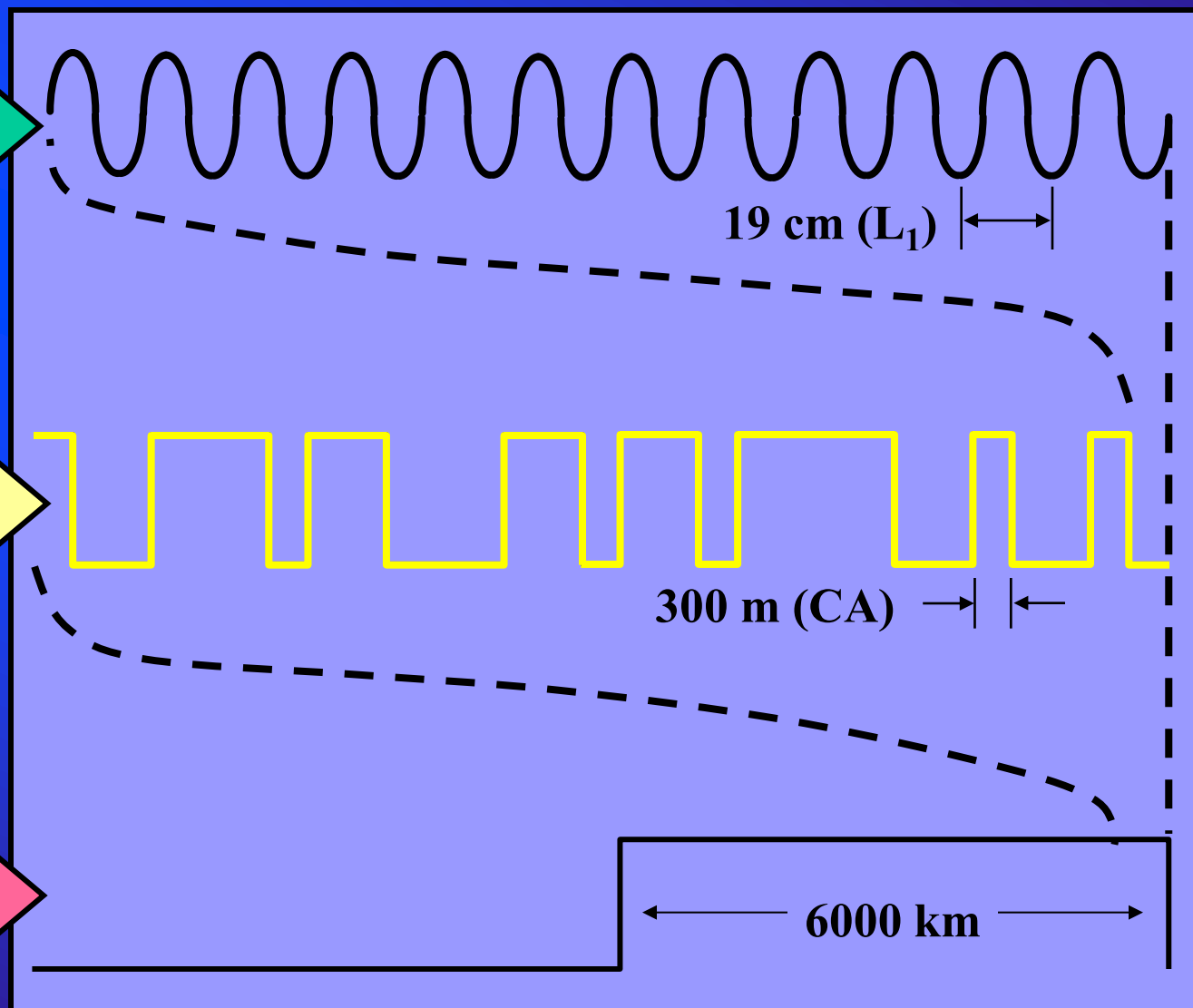
1575.42 MHz (L_1)
1227.60 MHz (L_2)

Code

1.023 Mcps (C/A),
period = 1 msec
10.23 Mcps (P(Y)),
period = 1 week

Navigation data

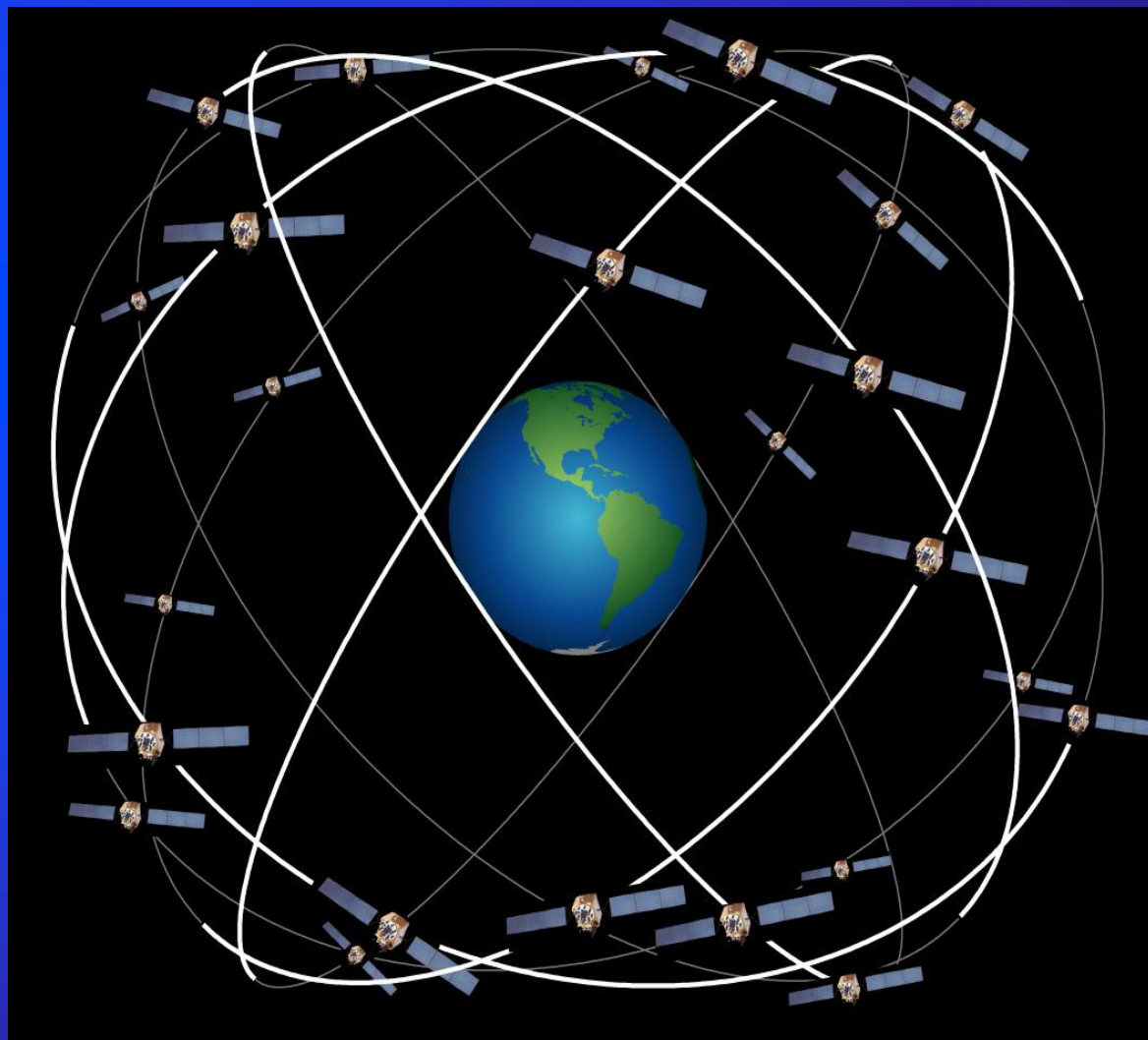
50 bits/second





The GPS Constellation

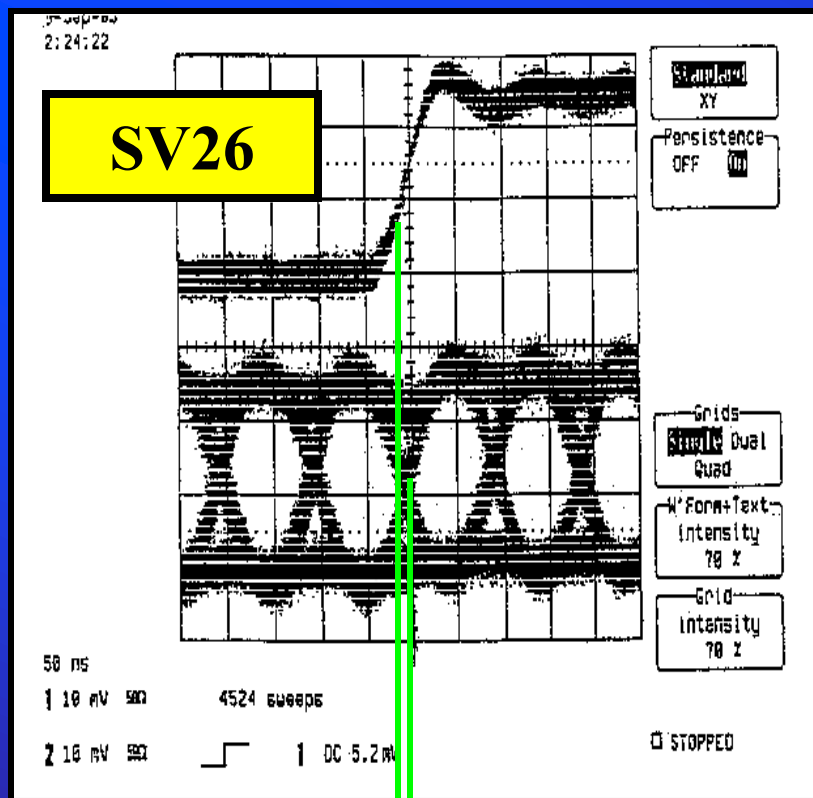
- GPS Block I
 - 02/78 – 10/85
 - 0 of 10 still on orbit
 - Prototypes for testing
- GPS Block II
 - 02/89 – 10/90
 - 3 of 9 still on orbit
- GPS Block IIA
 - 11/90 – 11/97
 - 17 of 19 still on orbit
- GPS Block IIR
 - 01/97 – present
 - 8 of 8 still on orbit



Courtesy Dr. Todd Walter



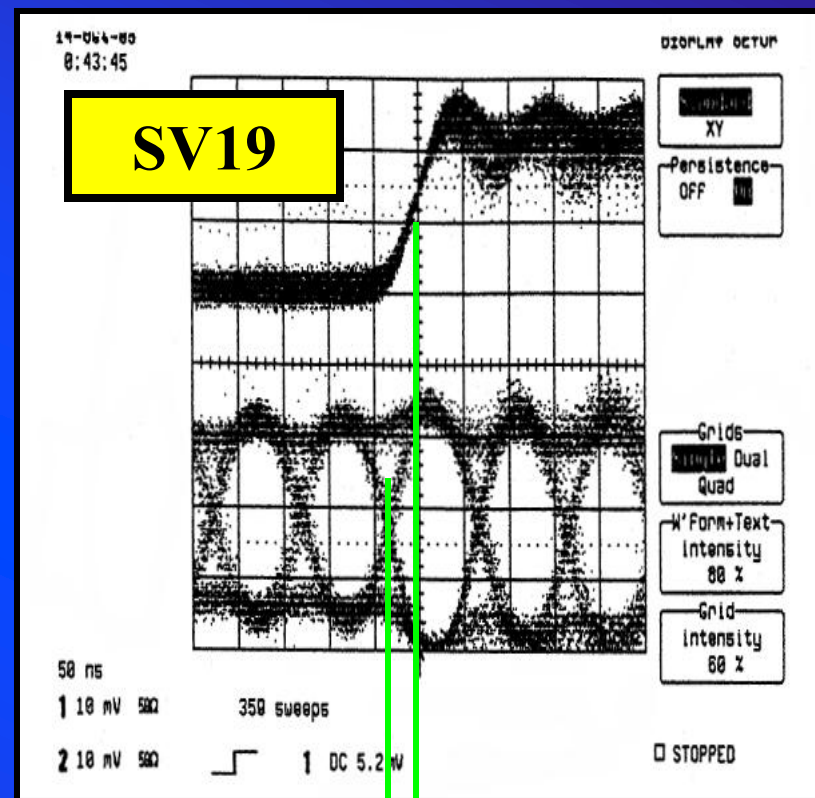
SV19 Signal (Time Domain)



5 nsec

- C/A code edges on healthy SV aligned to within 3-6 nsec*.

*equipment measurement accuracy



33 nsec

- C/A code edges on SV19 misaligned by ~ 28-33 nsec with respect to P code edges.

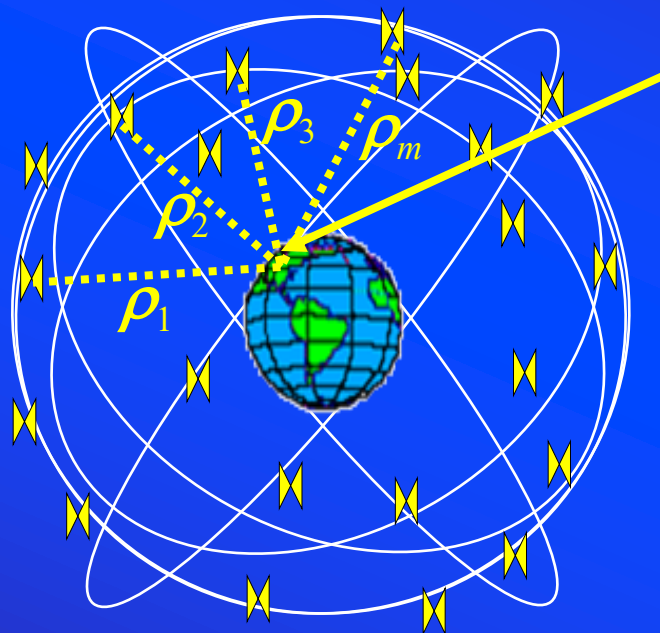


Contributions

- Geometric Analysis Of Original SV 19 Observations
 - Observed results (Oshkosh, WI) and waveforms (Camp Parks, CA)
 - Sensitivity of vertical error to PR error on SV 19
- Evil Waveforms
 - Theoretical analysis for “most evil” (worst) case [NTM 1999]
 - *Simplified ICAO threat space*
- Characterization of operational GPS Satellites
 - Spectral domain (SV19 revisited)
 - Correlation Domain (simulation using Arbitrary GPS Generator)
 - Time Domain [ION 2002, RTCA 2003]
- Tools for SQM Analysis
 - Arbitrary GPS Waveform Generator [ION 2000 Best-of-Session; *Navigation*]
 - Real-Time 48-Correlator Monitor Code [ION 2000]



Position Solution and DOP



•Initial user position and clock: $\begin{bmatrix} \mathbf{x}_0 & b_0 \end{bmatrix}^T$

•Line-of-sight (LOS) vector user $\rightarrow$ SVs:

$$\hat{\mathbf{L}}_m = -\begin{bmatrix} \cos(el_m) \sin(az_m) & \cos(el_m) \cos(az_m) & \sin(el_m) \end{bmatrix}^T$$

•The geometry matrix:

$$\mathbf{G} = \begin{bmatrix} \hat{\mathbf{L}}_1^T \\ \hat{\mathbf{L}}_2^T \\ \vdots \\ \hat{\mathbf{L}}_M^T \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

•Finding DOP:

$$\tilde{\mathbf{H}} = (\mathbf{G}^T \mathbf{G})^{-1} =$$

$$\begin{bmatrix} E^2 & \cdot & \cdot & \cdot \\ \cdot & N^2 & \cdot & \cdot \\ \cdot & \cdot & V^2 & \cdot \\ \cdot & \cdot & \cdot & T^2 \end{bmatrix}$$

•Iterative least-squares position solution:

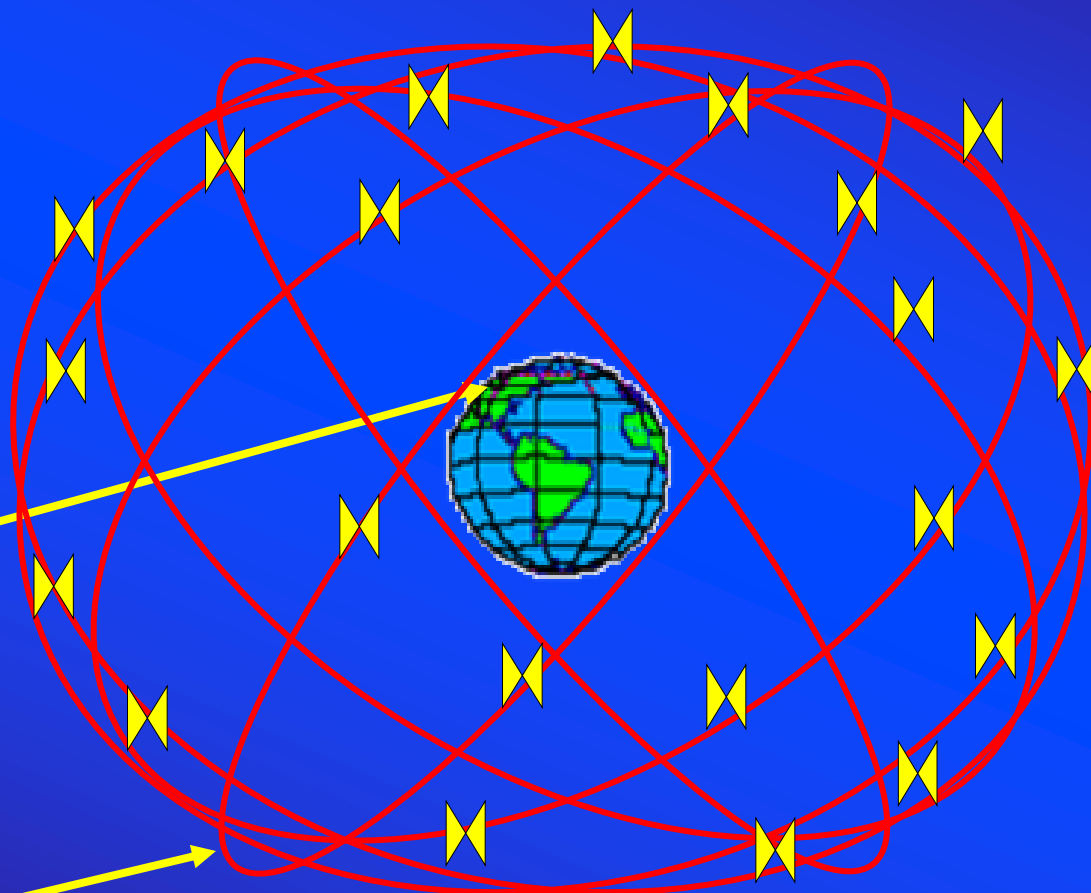
$$\begin{bmatrix} \hat{\mathbf{x}} \\ \hat{b} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_0 \\ b_0 \end{bmatrix} + (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \delta \mathbf{p}$$

$$\text{PDOP} =$$

$$\sqrt{E^2 + N^2 + V^2}$$



Geometric Analysis

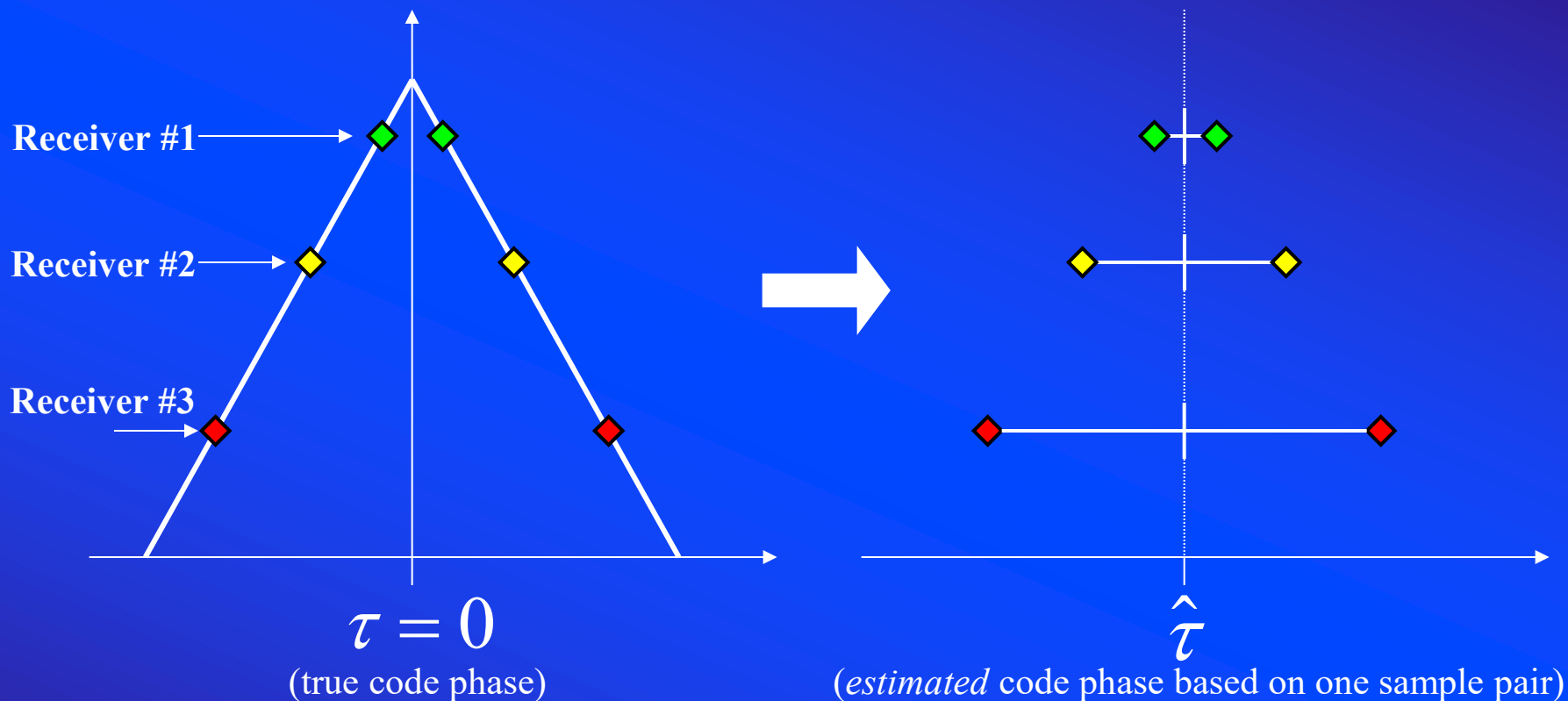


Location -
Oshkosh, WI
 $43^{\circ}59'45''$ N
 $88^{\circ}33'10''$ W

Almanac
GPS Week 704
(July 4→10, 1993)



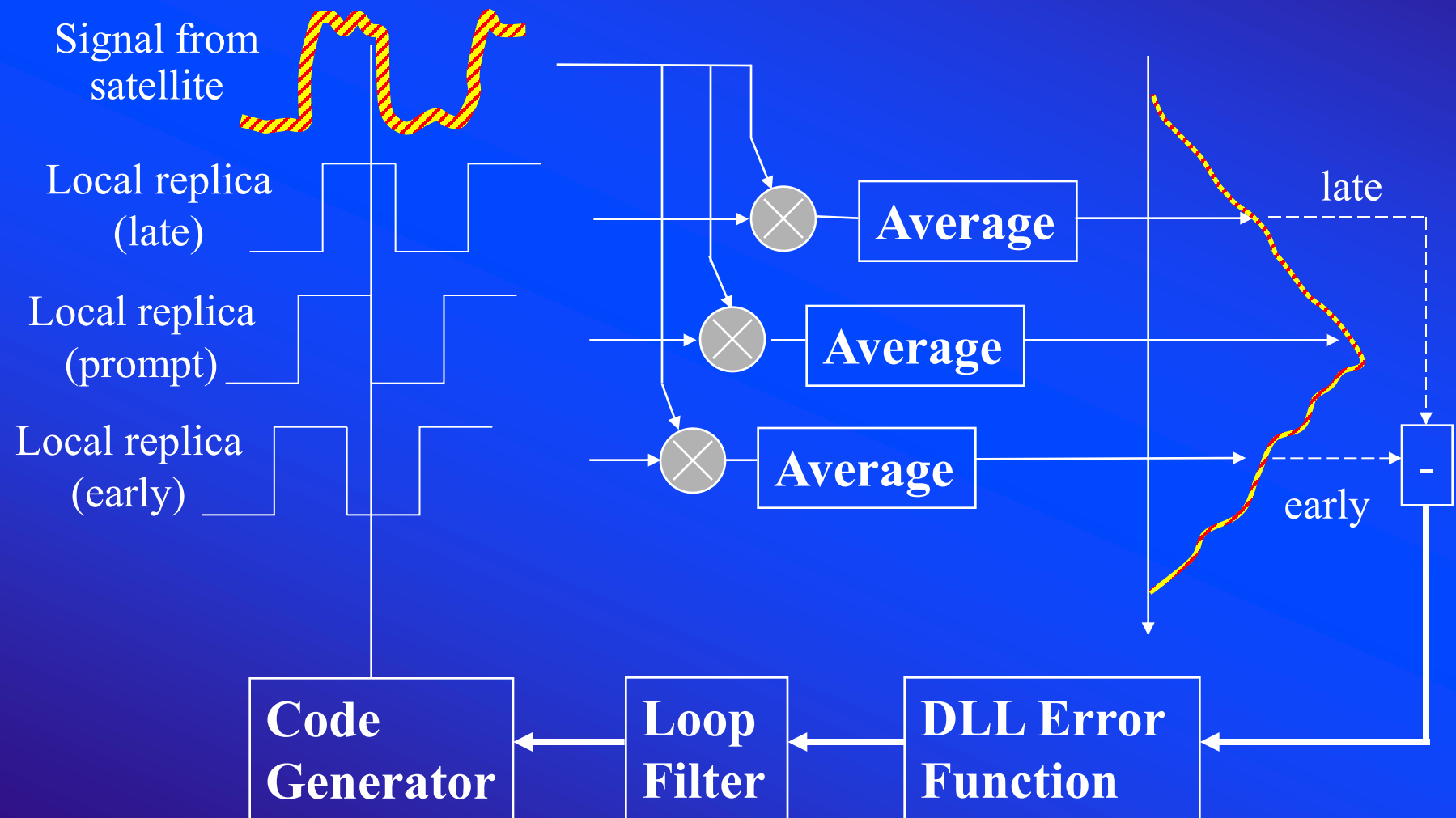
What a Receiver Really “Sees”



A receiver generally has no information about the complete structure of the underlying peak -- only the values of its correlator samples. (Here, all three receivers' correlator pairs make the same code phase estimate.)



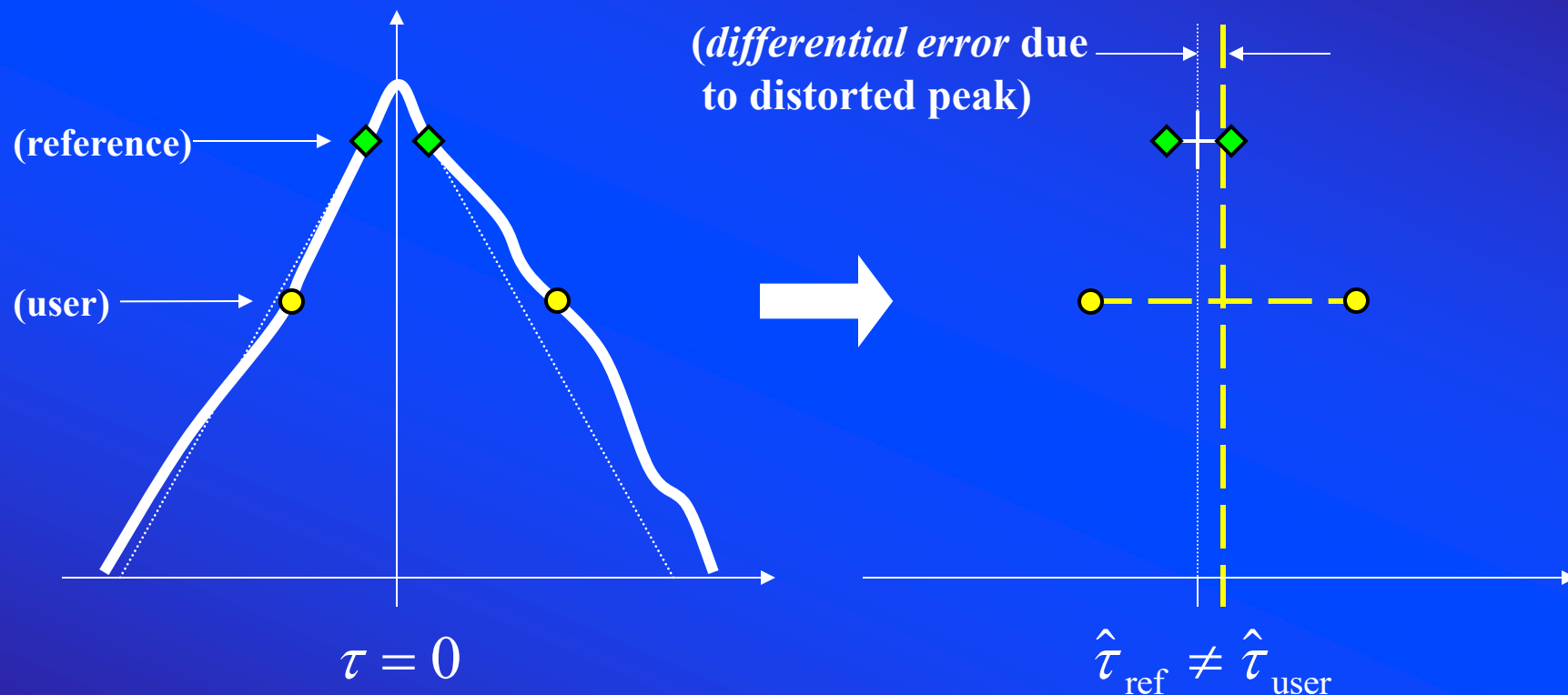
Distorted Code $\Rightarrow$ Distorted Peak





Differential Error

What happens when the peak is distorted?

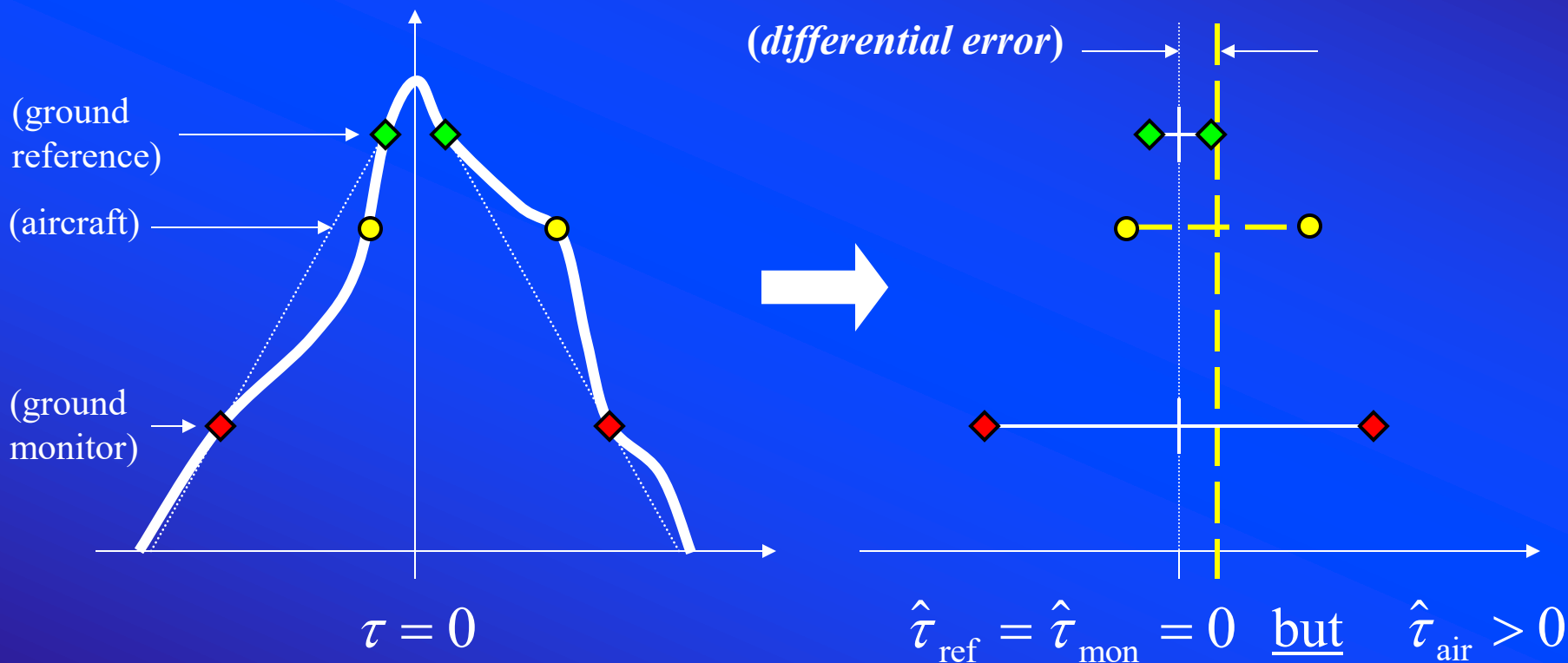


- Disagreement in code phase estimate between reference (ground station) and roving user is mapped onto a position error as a function of satellite geometry. Errors common to all SVs are mapped onto user clock error and corrected.



A Troublesome Example

Given one or more monitors, what errors can get through undetected?



- To the ground, this badly distorted peak is indistinguishable from the nominal correlation peak shown on the earlier slide!



Generalized Fault Model

- In general, a distorted GPS signal can be written as the sum of the nominal signal and a distortion term:

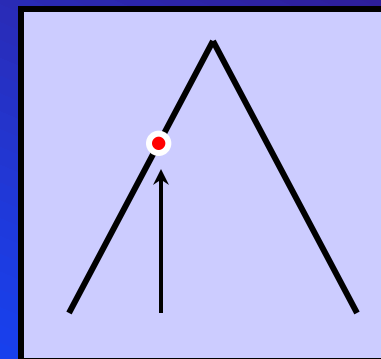
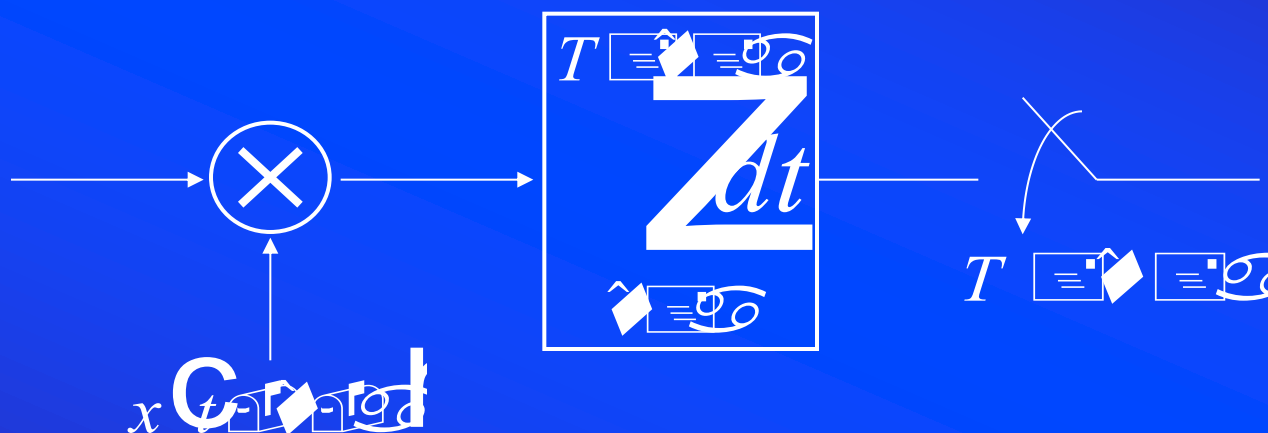
$$G(t) = \sqrt{2C_X} d(t - \tau) x(t - \tau) \cos(2\pi f_{L1} t + \theta) + j(t)$$

- Starting with this model, we can derive an explicit worst-case waveform, $j(t)$, as a function of the system parameters:
 - Number of monitor pairs and their spacings, $\{\alpha_1, \alpha_2, \dots, \alpha_M\}$
 - Ground filter bandwidth and shape, $B_G(f)$
 - Aircraft spacing, α_A
 - Aircraft filter bandwidth and shape, $B_A(f)$



Generalized Model: First Principles

A single correlator sample ...

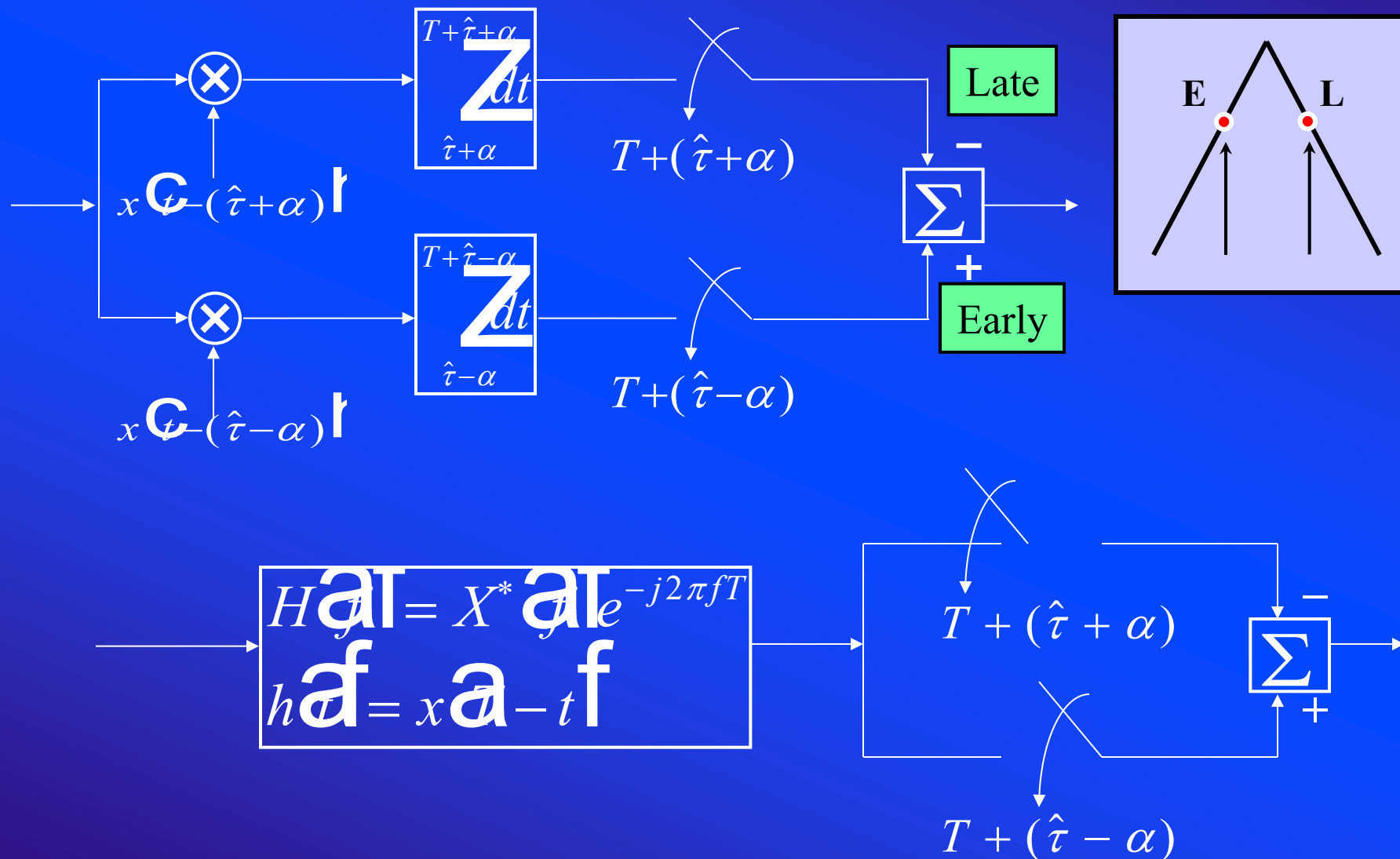


... and its frequency-domain equivalent



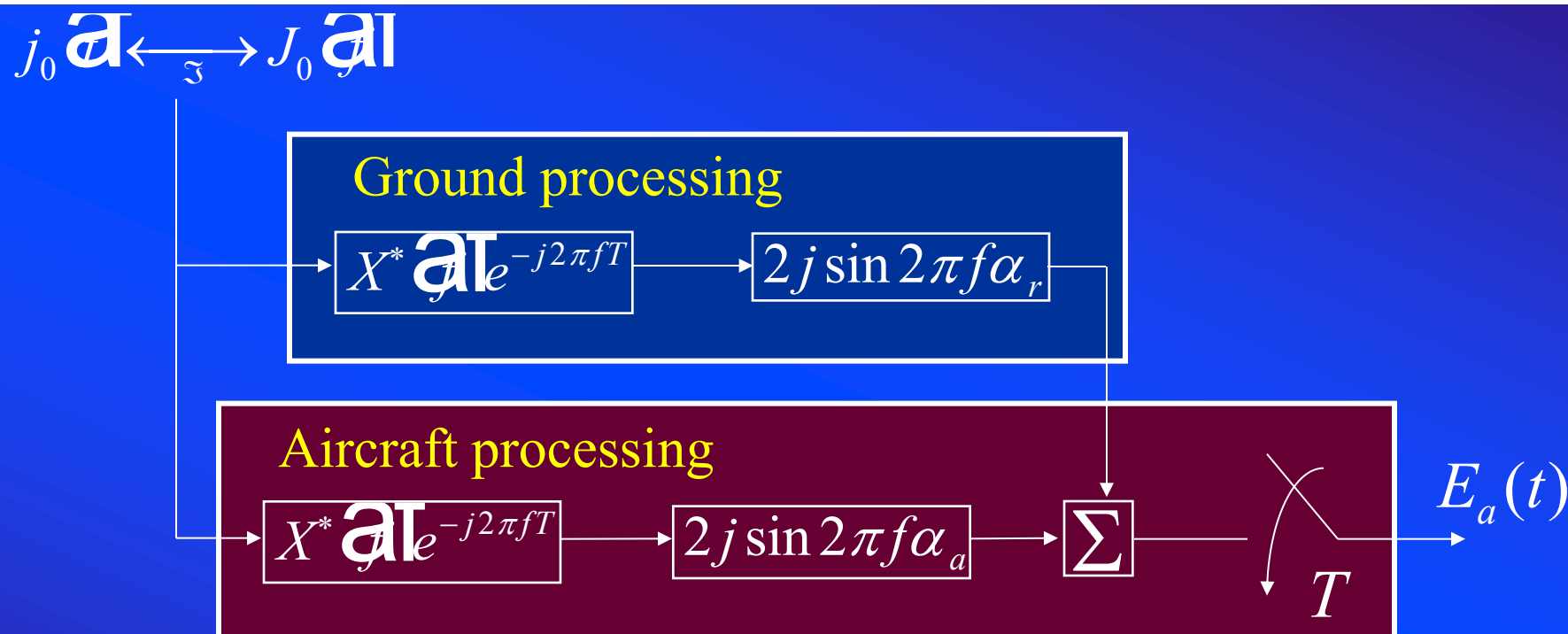


Generalized Model: Correlator Pair





Generalized Model: Basic LAAS (M=0)

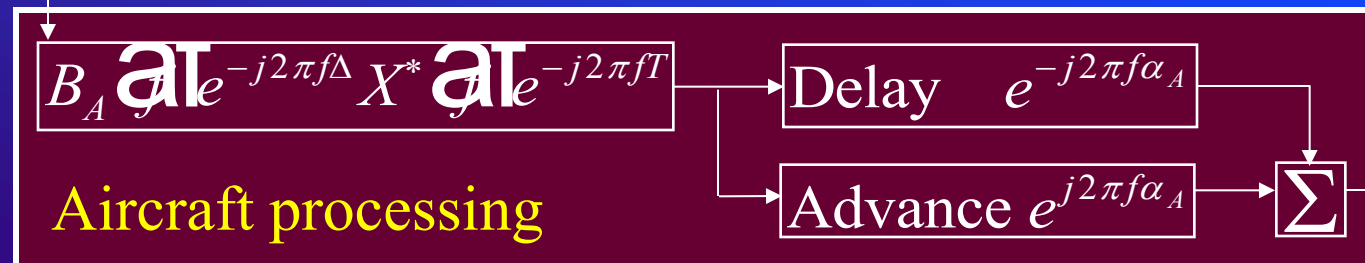
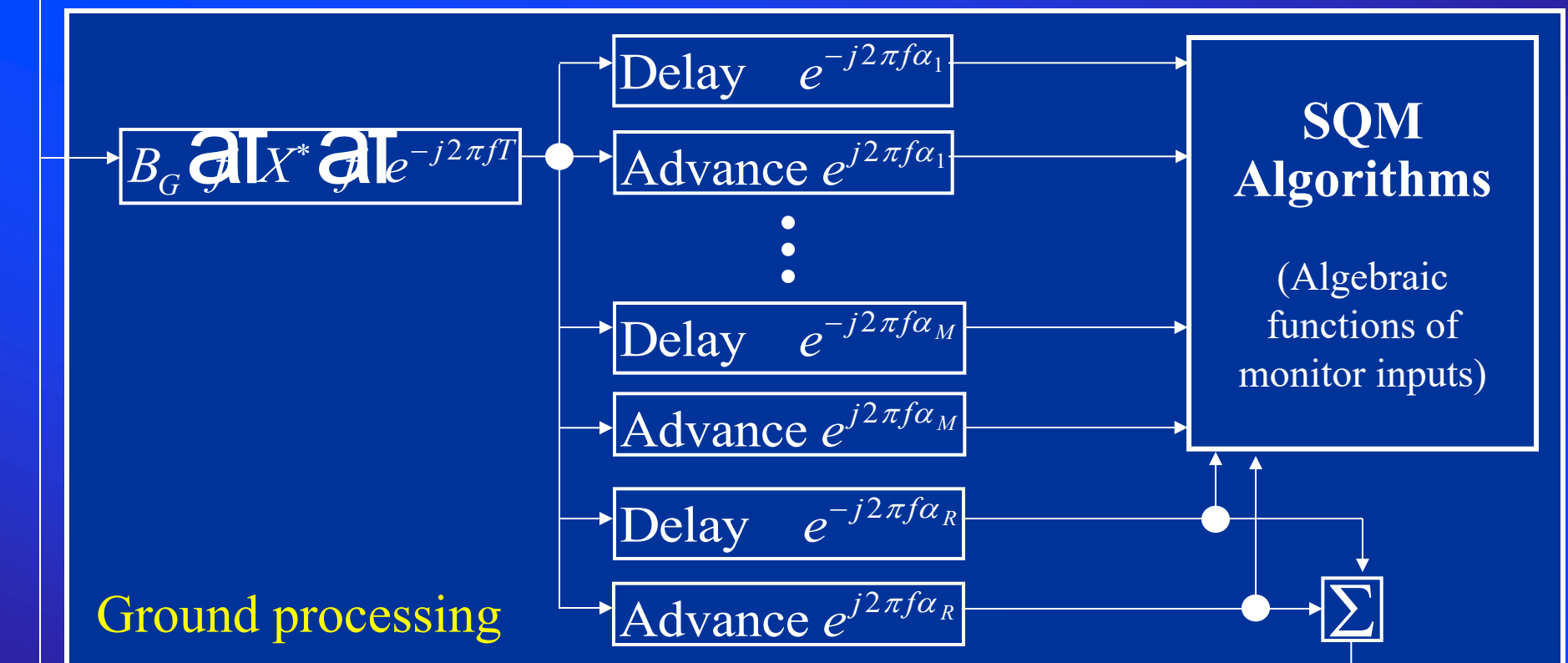


- The most evil waveform for M=0, $j_0(t)$, maximizes $E_a(t)$ and is matched to the overall transfer function.
- We can find the explicit form of $j_0(t)$ using Gram-Schmidt orthogonalization and the Schwartz inequality.



Generalized Model: Full LAAS ($M \geq 1$)

$$j_M \mathbf{a} \xleftrightarrow{\mathfrak{S}} j_M \mathbf{a}$$



$$\Sigma \rightarrow E_a \mathbf{a}$$



The Most Evil Waveform (1)

The most evil waveform for M monitors, J_M , lies in the null space of

$$\mathbf{N}_m(f) = X^*(f)(\sin 2\pi\alpha_m f - \sin 2\pi\alpha_r f) \mathbf{S}_{m=1}^M$$

and maximizes $E_a(t) = \langle J_M, Y_a \rangle$. We can construct such a function using the Schwartz inequality and Gram-Schmidt orthogonalization.

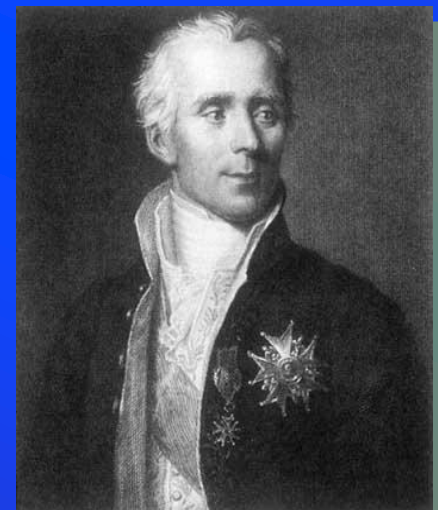
For example, the $M=2$ case is:

$$J_2(f) = Y_a^*(f) - k_1 Y_1(f) - k_2 Y_2(f)$$



With

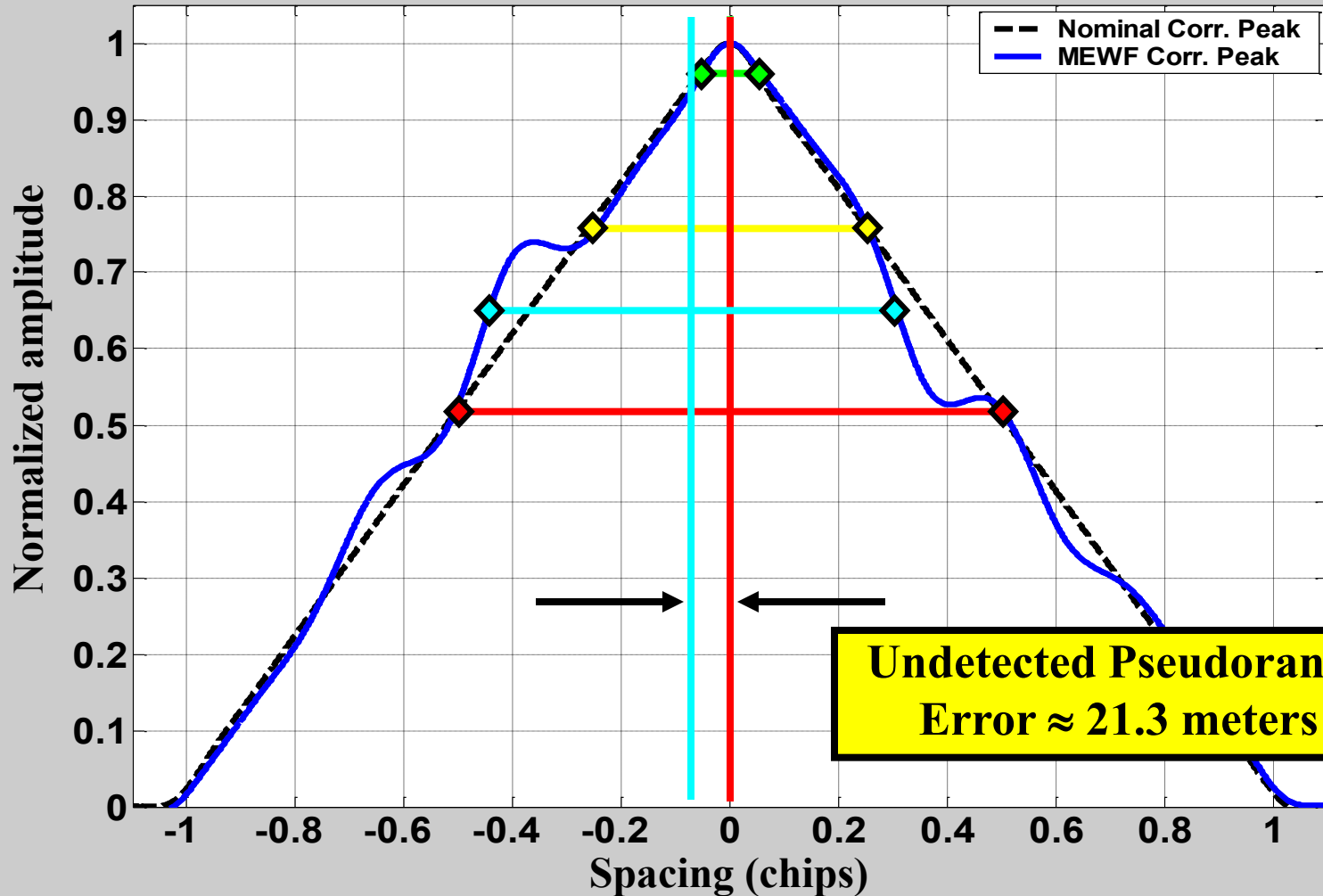
$$\langle J_2, Y_1 \rangle = 0 \quad \text{and} \quad \langle J_2, Y_2 \rangle = 0$$





The Most Evil Waveform (3)

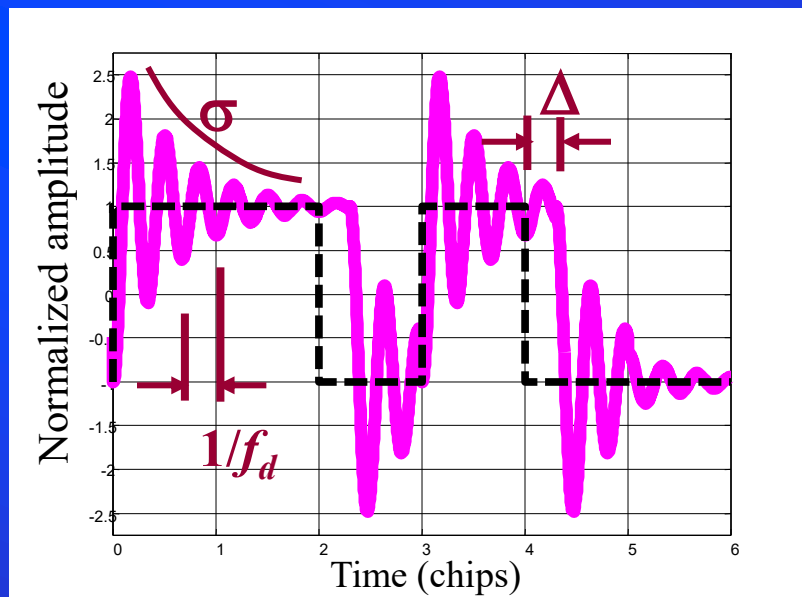
$\alpha_{\text{Ref}} = 0.1$, $\alpha_{\text{Mons}} = \{0.5, 1.0\}$, $\alpha_{\text{Air}} = 0.75$ 16MHz BPFs S/J = +6dB



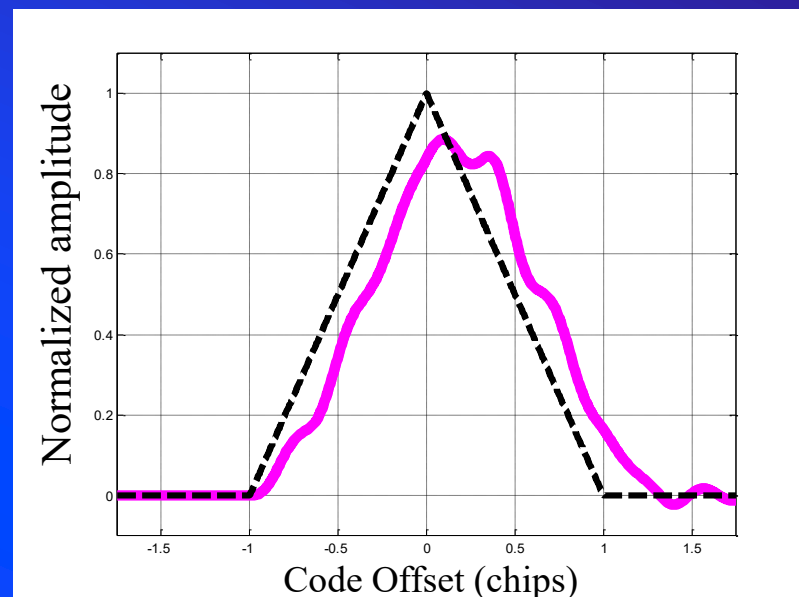
Undetected Pseudorange
Error ≈ 21.3 meters



ICAO Threat Model (1)



- “Analog” failure component
 - Two-pole linear system (LHP)
 - Completely described by σ and f_d
- “Digital” failure component
 - Delay/advance in falling (or rising) edges only
 - Completely described by Δ

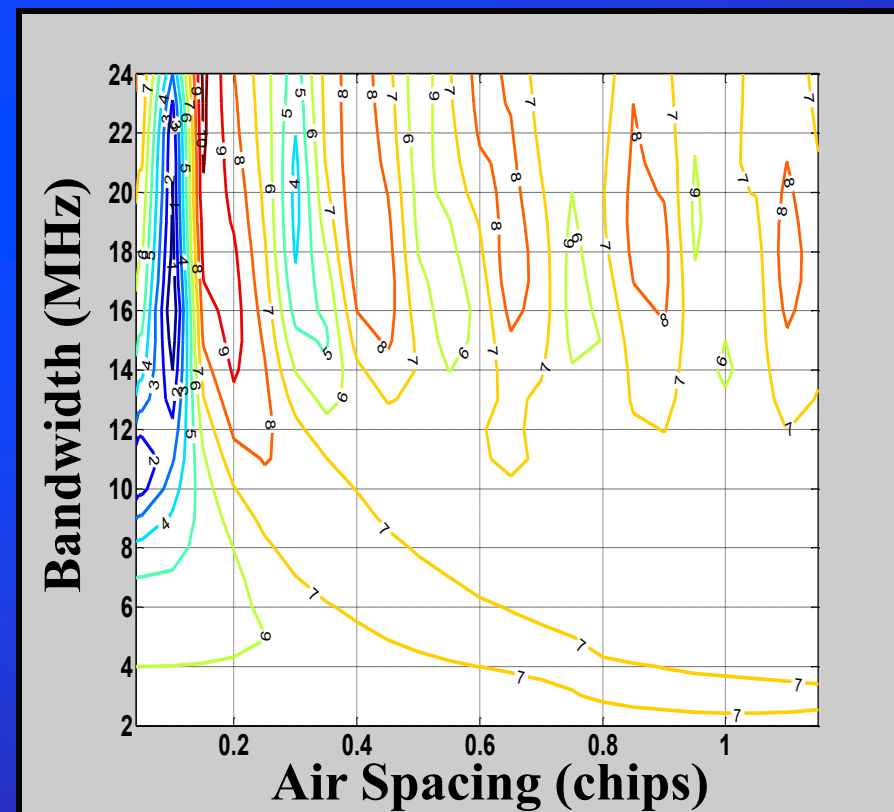


- Encapsulates essential pathologies hypothesized for SV19
 - side-to-side asymmetry
 - false peaks
 - flat top
- Consistent with empirical evidence (ringing, edge lag)
- Bounded and physically realizable



ICAO Threat Model (2)

- Specifies a class of signal anomalies which could lead to hazardously misleading information (HMI)
- Implies a dominant error source (single complex pole pair)
- Defines a three-dimensional threat space
 - Extensively analyzed by Phelts *et. al.* (2001)
 - Maximum vertical error for a particular correlator spacing and front-end BW is found by exploring this space





ICAO Model: Application (1)

- This model is useful for detailed analysis of signal anomalies. For example, consider what it tells us about the troublesome spikes seen in the spectrum of SV19:



- How does a non-zero Δ affect the spectrum? To see its effects, consider the *difference* waveform:

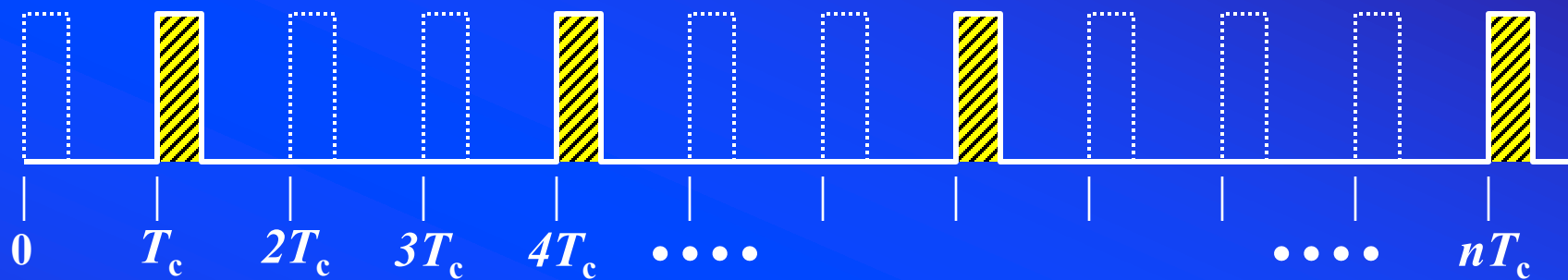


- Periodic in $T_{\text{epoch}} = 1$ msec; exactly 256 Δ -wide pulses at nT_{chip}



ICAO Model: Application (2)

- This is *almost* periodic in T_{chip} :



- Compute the Fourier Series expansion starting with the fundamental frequency

$$f_0 = 1/T_{\text{chip}} = 1.023 \text{ MHz}$$

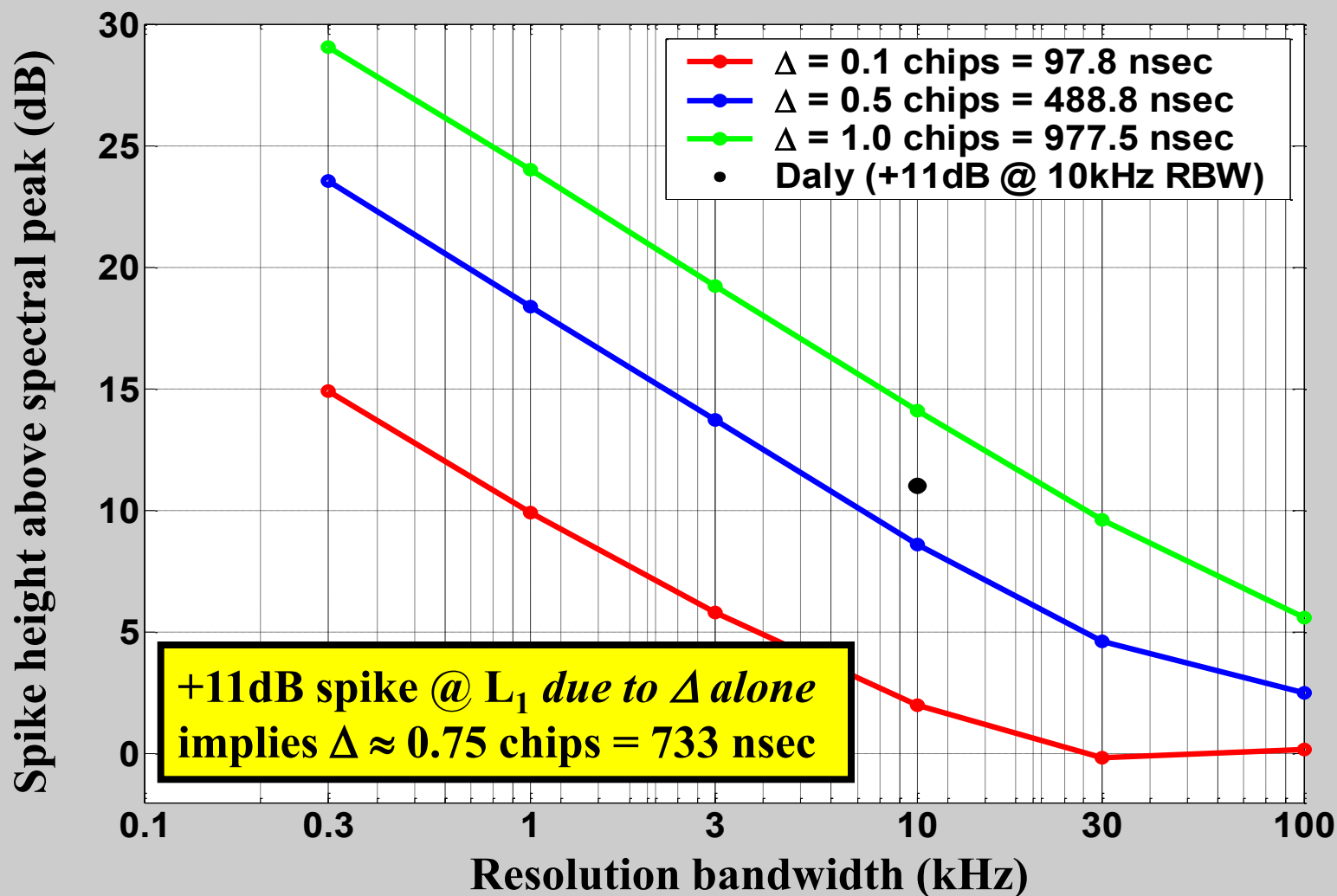
and scaling the resulting coefficients by the factor

$$k = 256/1023 \approx 0.25$$

- This gives us an explicit relationship between Δ and the amplitude of spectral spikes at $f = L_1 + nf_0$ (C/A nulls).

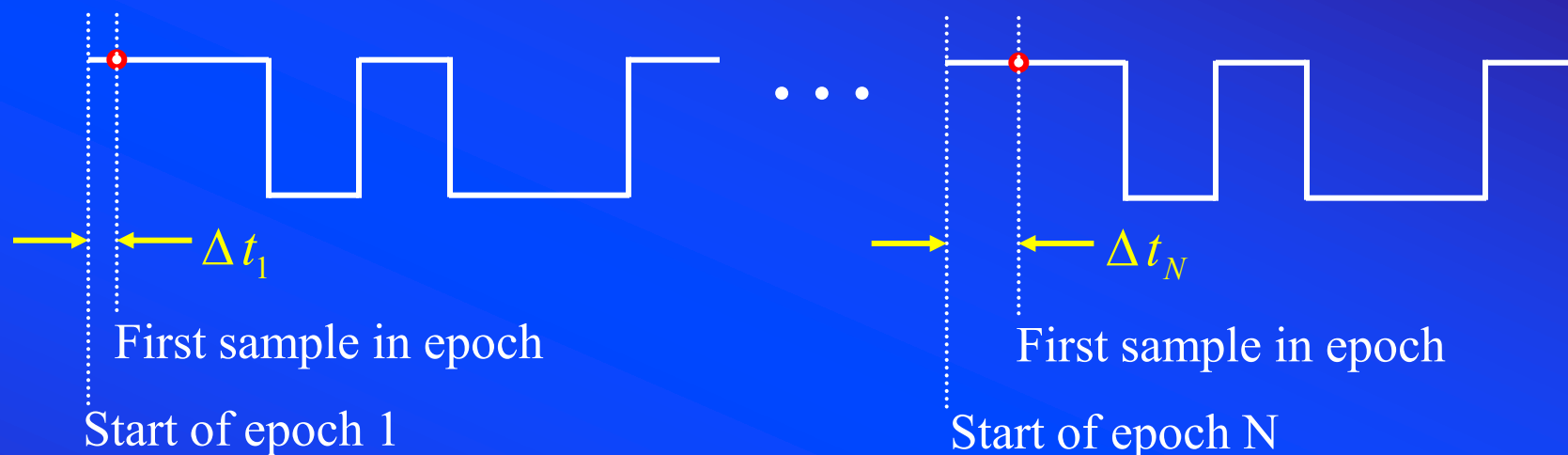


ICAO Model: Application (3)





Synchronous Sampling (1)

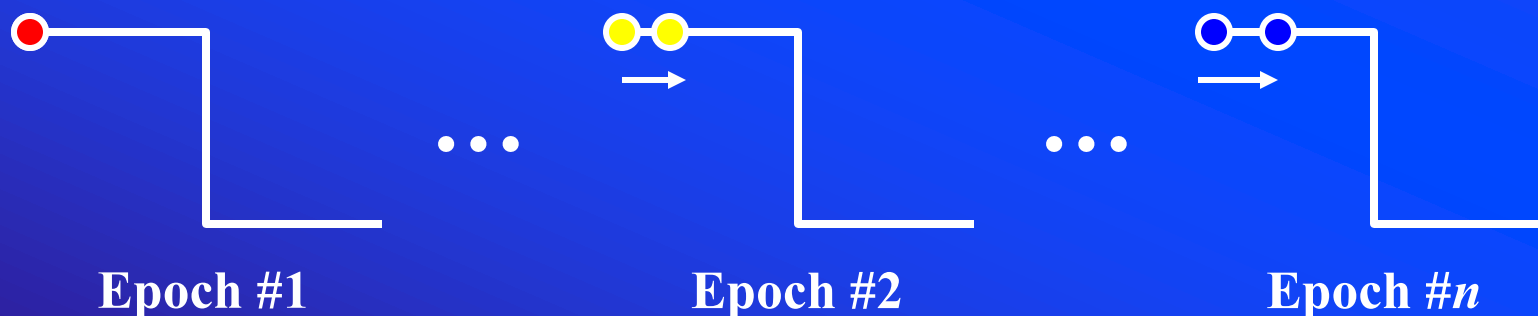


- Uncorrected errors (Doppler, LO, ADC clock) lead to sample “drift” with respect to underlying features
 - 47.5 MHz, 250 msec, 2 kHz Doppler (uncompensated) $\Rightarrow$
Accumulated error = $\Delta t_N - \Delta t_1 \approx \underline{15.08 \text{ samples}} \approx \underline{0.32 \text{ chips}}$
- Tight control of timebase required to avoid this error.



Vernier Sampling (1)

- Given...
 - An accurate local oscillator (Rb standard accurate to 1mHz at 10MHz);
 - A method to reliably predict Doppler shift of a given satellite as seen from a known location at a given time (demonstrated as part of Oshkosh SV19 analysis);
 - A/D converter with sample rate adjustable in fine steps (VSA specified to 1mHz nominally)
- What happens when we *reduce* the sample rate?





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 - An accurate local oscillator (Rb standard accurate to 1mHz at 10MHz);
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 - A/D converter with sample rate adjustable in fine steps (VSA specified to 1mHz nominally)
- What happens when we *reduce* the sample rate?
- We can increase the effective sample rate by a factor of n over the maximum instrument sample rate
 - This occurs at the expense of noise averaging
 - Affords improved time resolution provided data set is long enough

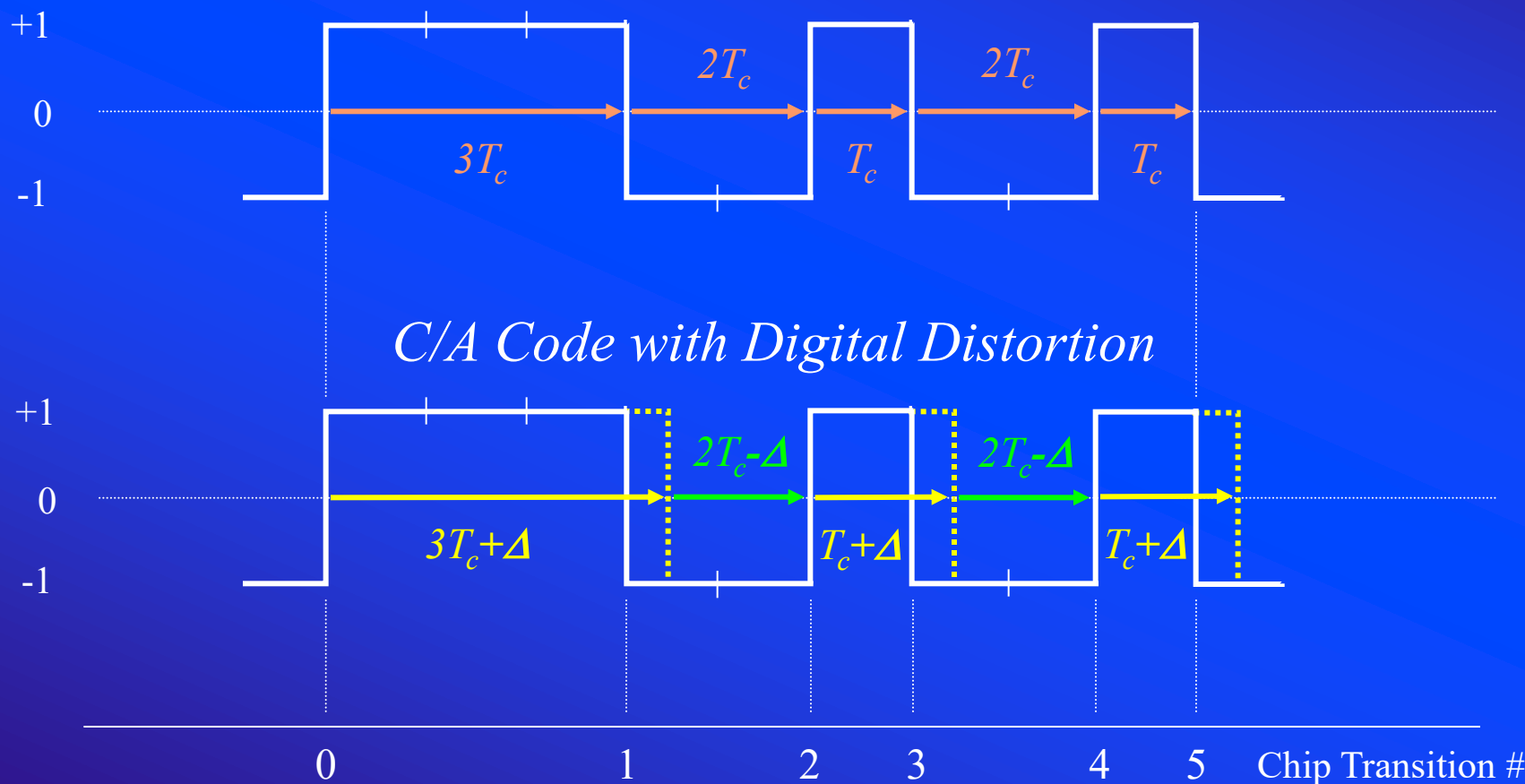


Aggregate Epoch



Measuring Digital Distortion (1)

Ideal C/A Code





Summary of Aviation Requirements

SPS/RAIM
+ INS

WAAS

LAAS
(satisfies
WAAS
reqmts.
within
VDB
coverage)

Phase of Flight	Accuracy (95% Error)	Integrity			Continuity Pr(loss of navigation)	Availability	
		Time to Alert	Alert Limit	Pr(MI)		Threshold	Objective
Oceanic Enroute	H: 12.4 nmi	2 min	H: 12.4 nmi	10^{-7} / hour	10^{-5} / hour	0.99	0.999 – 0.99999
Domestic Enroute	H: 2.0 nmi	1 min	H: 2.0 nmi	10^{-7} / hour	10^{-6} / hour	0.99	0.99999
Terminal Area	H: 0.4 nmi	30 sec	H: 1.0 nmi	10^{-7} / hour	10^{-6} / hour	0.99	0.99999
Non-prec. Approach	H: 220 m	10 sec	H: 556 m	10^{-7} / hour	10^{-5} / hour	0.99	0.99999
LNAV/ VNAV	H: 220 m	10 sec	H: 556 m V: 50 m	10^{-7} / hour	5.5×10^{-5} / approach	0.99	0.99999
LPV (APV 1.5)	H: 16 m V: 20 m	10 sec	H: 40 m V: 50 m	2×10^{-7} / approach	5.5×10^{-5} / approach	0.99	0.99999
APV-2	H: 16 m V: 7.6 m	6 sec	H: 40 m V: 20 m	2×10^{-7} / approach	5.5×10^{-5} / approach	0.99	0.99999
Cat. I Prec. Appch.	H: 16 m V: 4? 7.6 m	6 sec	L: 40 m V: 10? 12 m	2×10^{-7} / approach	5.5×10^{-5} / approach	0.99	0.99999
Cat. II Prec. Appch.	H: 6.9 m V: 2.0 m	2 sec	L: 17.3 m V: 5.3 m	2×10^{-9} / approach	4×10^{-6} / 15 sec	0.99	0.99999
Cat. III Precision Appch.	H: 6.1 m V: 2.0 m	1 – 2 sec	L: 15.5 m V: 5.3 m	2×10^{-9} / approach	L: 2×10^{-6} / 30 sec V: 2×10^{-6} / 15 sec	0.99	0.99999

Being reconsidered by RTCA

Original Source: *GPS Risk Assessment Study: Final Report.* Johns Hopkins University Applied Physics Laboratory, VS-99-007, January 1999. <http://www.jhuapl.edu/transportation/aviation/gps/>